FEA Membranes

Validation Manual

Element library, constraint methods and stress-intensity extraction

Southward Consulting Ltd

12 June 2026 Rev. A (draft — NASTRAN comparison columns to be populated)

Summary

This manual records the verification and validation of the *FEA Membranes* native application: a 2D plane-stress finite element tool with bilinear (Quad4) and quadratic serendipity (Quad8) membrane elements, axial bar elements, XY-decoupled spring (fastener) elements, RBE2 rigid constraints, quarter-point crack-tip modelling and stress-intensity-factor extraction by domain interaction integral.

Every case in this manual is implemented as an automated test executing on every build of the software (46 cases at this revision, all passing). The benchmark models are regenerated by the test suite and stored alongside this document; the figures are rendered from those models by the application's batch renderer. **Every number and figure in this manual is therefore reproducible from the committed code with two commands** (Appendix A).

Headline results:

- Closed-form element cases (uniaxial patch, pure bending, bar, spring, enforced displacement, RBE2 ties): **exact** to the stated tolerances (typically 10^{-6} or better, relative).
- MacNeal & Harder straight cantilever, 6 elements: ratio **0.987** (rectangular), **0.991–0.997** (parallelogram), **0.967–0.982** (trapezoidal), converging to 0.9993 under refinement.
- Stress intensity factors vs handbook solutions: $-0.1\,\%$ (edge crack and centre crack), domain-independent to $< 1\,\%$.

Columns are provided throughout for comparison against MSC NASTRAN CQUAD8(R) results, to be populated as those runs are completed.

Contents

Chapter 1

The software and its methods

1.1 Scope

FEA Membranes is a Windows-native linear-static finite element application for 2D plane-stress idealisations of sheet structures: skins, doublers, stringers and fastened joints, with crack-tip modelling for damage-tolerance work. It is developed and maintained by Southward Consulting Ltd; source, tests, benchmark models and this manual live in one repository.

1.2 Element library

Quad4 Isoparametric 4-node bilinear membrane, plane stress, 2×2 Gauss integration.

Quad8 Isoparametric 8-node serendipity membrane, plane stress, 2×2 **reduced integration** — the standard scheme for this element (MSC NASTRAN QUAD8, Abaqus CPS8R). Full 3×3 integration was measured to carry $\approx 0.5\,\%$ additional stiffness on MacNeal's cantilever (Section 3) and was rejected. The single spurious zero-energy mode of an isolated reduced-integration Q8 is non-communicable in meshes and additionally guarded by the solver's residual check (Section 7).

Bar 2-node axial rod, $k = EA/L$, arbitrary orientation. No bending or transverse stiffness (by design); tension positive.

Spring 2-node fastener idealisation with *independent* stiffness k in global X and Y. Deliberately **not** an axial spring: this is the classic sheet load-transfer fastener model, normally used between coincident nodes.

RBE2 Nastran-style rigid element tying dependent nodes' X and/or Y translations to an independent node. Enforced as an *exact* multipoint constraint by DOF merging — not a penalty.

1.3 Solution method

Sparse direct solution (CSparse LU with minimum-degree ordering). Fixed and enforced-displacement boundary conditions are applied by *direct elimination* (partitioned solve), so prescribed values are satisfied to machine precision with no penalty-number conditioning. RBE2 ties merge DOFs into shared solve unknowns before assembly. Every solution is checked against the equilibrium residual $\|Ku - f\| \leq 10^{-6}\|f\|$; models with rigid-body mechanisms are rejected with a diagnostic rather than returning silently wrong answers.

1.4 Stress recovery

Element stresses are reported at the element centre. Nodal stresses are recovered by evaluating each element's stresses at its node locations and averaging over the elements attached to each node; von Mises is formed from the averaged components. The contour display interpolates the averaged nodal values (Gouraud shading); a per-element (unaveraged) display is also available.

1.5 Crack modelling and stress intensity factors

A crack is introduced by splitting the mesh along a straight path of element edges: every non-tip path node is duplicated so the two faces can separate, and the midside nodes on all edges emanating from the tip are moved to the quarter points (Barsoum), embedding the $1/\sqrt{r}$ singularity through the isoparametric mapping. Cracks require a Quad8 mesh.

K_I and K_{II} are extracted by a **domain interaction integral** (equivalent domain integral form):

$$I^{(1,2)} = \int_A \left[\sigma_{ij}^{(1)}\,\partial_1 u_i^{(2)} + \sigma_{ij}^{(2)}\,\partial_1 u_i^{(1)} - W^{(1,2)}\delta_{1j}\right]\partial_j q\, dA, \qquad K = \tfrac{E'}{2}\, I, \tag{1.1}$$

with Williams auxiliary fields of unit K_I or K_{II}, the weight q ramping radially from 1 at the tip to 0 at the domain boundary (≈ 3 tip-element lengths), and all quantities in the crack-local frame. A two-point quarter-point displacement-correlation estimate is reported alongside as a cross-check; a large divergence between the two flags a suspect tip mesh.

Chapter 2

Closed-form element verification

The cases in this chapter have exact answers for the discretised problem; the elements must reproduce them to tight tolerances. They are patch-test class *verification* cases.

2.1 V1 — Quad4 uniaxial tension

Single 10×10 element, $E = 10^7$, $\nu = 0$, $t = 0.1$, total 1000 lb on the free edge (consistent loads). Closed form: $\sigma_x = 1000$ psi, $u = 1.0 \times 10^{-3}$ in.

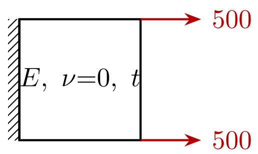


Figure 2.1: V1 setup.

Quantity	Reference	FEA Membranes	Error	NASTRAN CQUAD4
u at free edge (in)	1.000×10^{-3}	1.000×10^{-3}	$< 10^{-6}$	—
σ_x (psi)	1000	1000	$< 10^{-6}$	—
σ_{VM} (psi)	1000	1000	$< 10^{-6}$	—
ΣR_x (lb)	-1000	-1000	$< 10^{-6}$	—

Table 2.1: V1 results (test `Quad_UniaxialTension_ExactForBilinear`).

2.2 V2 — Quad8 uniaxial patch test

Single 8-node element, same plate as V1, consistent quadratic-edge loads ($P/6$, $4P/6$, $P/6$). The application's *Distributed Load* command generates this distribution automatically and is itself verified to reproduce the exact field (tests `DistributedLoad_*`).

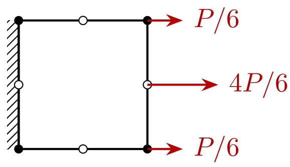


Figure 2.2: V2 setup: Quad8 patch test (open circles are midside nodes) with consistent quadratic-edge loads.

Quantity	Reference	FEA Membranes	Error	NASTRAN CQUAD8
u at all free-edge nodes	1.000×10^{-3}	1.000×10^{-3}	$< 10^{-9}$	—
σ_x at all 8 averaged nodes	1000	1000	$< 10^{-4}$	—
ΣR_x (lb)	-1000	-1000	$< 10^{-5}$	—

Table 2.2: V2 results (test `Quad8_PatchTest_UniaxialTensionExact`).

2.3 V3 — Quad8 pure in-plane bending (single element)

A single Q8 (20×10) under pure bending applied as a consistent linear end traction $\sigma_x(y) = 200\,y$. The quadratic displacement field carries linear strain exactly — the defining capability the bilinear element lacks.

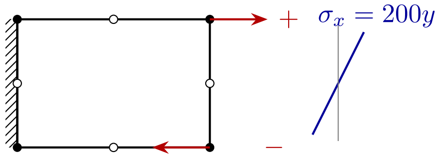
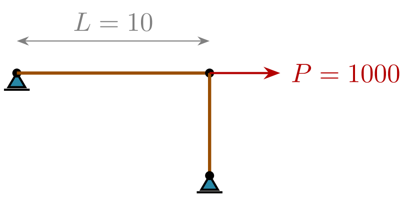


Figure 2.3: V3 setup: single Quad8 in pure bending; the recovered stress profile is exactly linear.

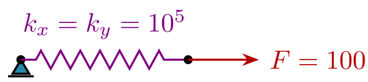
Quantity	Reference	FEA Membranes	Error	NASTRAN CQUAD8
σ_x at every node	$200\,y$	$200\,y$	$< 10^{-3}$	—

Table 2.3: V3 results (test `Quad8_PureBending_ExactLinearStress`).

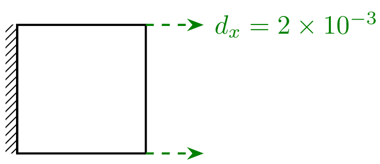
2.4 V4 — Bar, spring, enforced displacement, RBE2



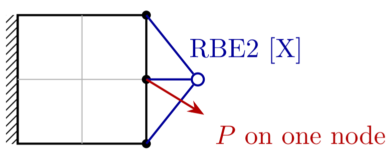
(a) Bar: axial elongation PL/EA.



(b) Spring: decoupled $F = k\delta$ per direction.



(c) Enforced edge displacement (dashed green).



(d) RBE2 spider tying the edge in X.

Figure 2.4: V4 setups: (a) axial bar with transverse support bar; (b) XY-decoupled fastener spring; (c) enforced displacement by direct elimination; (d) RBE2 exact multipoint constraint with the load on one node of the tied group.

Case	Closed form	FEA Membranes	Error	NASTRAN
Bar elongation	$\delta = PL/EA = 0.01$	0.01	$< 10^{-9}$	—
Bar force / stress	P=1000 (T), σ=10^4	exact	$< 10^{-6}$	—
Spring $F = k\delta$	$\delta = 10^{-3}$	exact	$< 10^{-9}$	—
Enforced displacement	$d_x = 2 \times 10^{-3}$ exactly	exact	$< 10^{-12}$	—
RBE2 tied d_x (X-only tie)	identical across group	identical	$< 10^{-12}$	—
RBE2 group δ via bar	PL/EA	exact	$< 10^{-9}$	—

Table 2.4: V4 results (tests `Bar_AxialElongation_PLOverEA`, `Spring_DecoupledXY_ForceAndDisplacement`, `EnforcedDisplacement_IsExact`, `Rbe2_*`).

2.5 V5 — Stress recovery (nodal averaging)

A constant stress field survives corner extrapolation and nodal averaging unchanged at every node (exact to 4 decimals). A deliberate thickness step between two elements (σ_x = 500 / 1000 psi) reports exactly the mean, 750 psi, at the shared interface nodes (tests `NodalStresses_*`).

Chapter 3

MacNeal & Harder cantilever

The straight cantilever of MacNeal & Harder (1985): $6.0 \times 0.2 \times 0.1$, $E = 10^7$, $\nu = 0.3$, unit in-plane shear load at the tip, 6 elements, reference tip deflection (including shear) $\delta_{ref} = 0.1081$. The distorted-element variants are built exactly as a user would build them: six surfaces sharing corner points, one Quad8 per surface, seams stitched with the coincident-node merge. Model files: `macneal-*.json`.

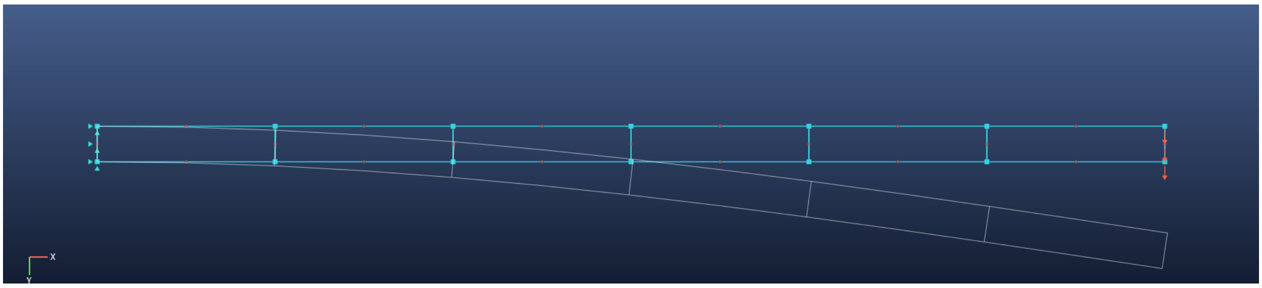


Figure 3.1: Rectangular mesh, deformed shape overlaid (white).

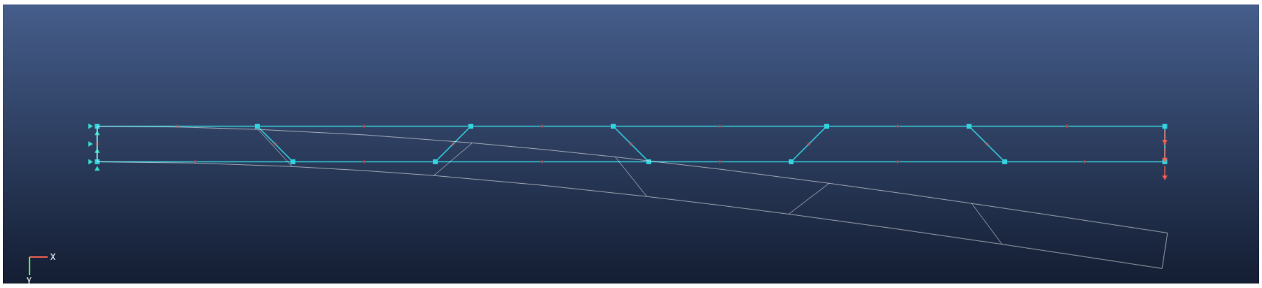


Figure 3.2: Trapezoidal mesh, 45° edge angle, deformed shape overlaid.

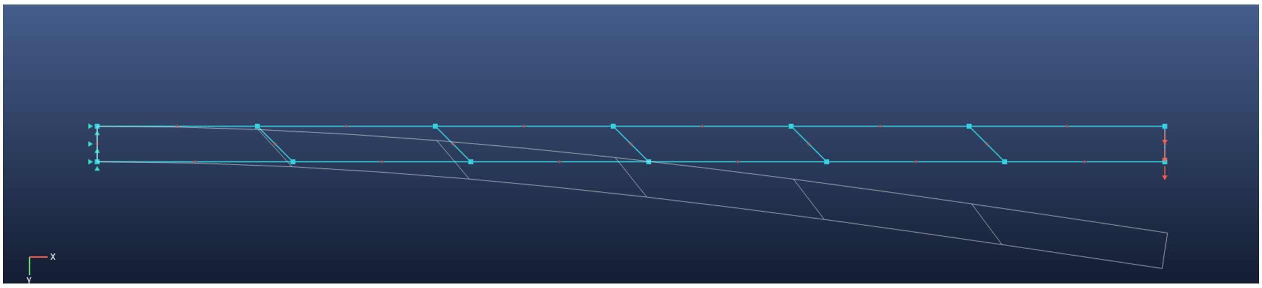


Figure 3.3: Parallelogram (skew) mesh, 45°, deformed shape overlaid.

Mesh (6 elements)	δ/δ_{ref}	Error	NASTRAN CQUAD8R	Notes
Rectangular	0.9868	$-1.32\,\%$	—	
Parallelogram 30°	0.9914	$-0.86\,\%$	—	affine mapping preserved
Parallelogram 45°	0.9969	$-0.31\,\%$	—	
Trapezoid 30°	0.9816	$-1.84\,\%$	—	
Trapezoid 45°	0.9666	$-3.34\,\%$	—	graceful, no locking

Table 3.1: Distorted-shape results (tests `Quad8_MacNealCantilever_*`).

Mesh	6×1	12×1	24×2	48×4	96×8
δ/δ_{ref}	0.9868	0.9933	0.9984	0.9992	0.9993

Table 3.2: Mesh convergence, rectangular elements. The converged residual ($-0.07\,\%$) is the physical fully-clamped-root effect relative to beam theory, not element error.

Chapter 4

Plate far-field and system-level cases

4.1 P1 — Clamped plate far-field stress

100×100 plate, 20×20 Quad4 mesh, $\nu = 0.3$, clamped edge, uniform end tension. Saint-Venant far-field $\sigma_x = 210$ psi at mid-plate: measured within 2 % (test `Plate_20x20_FarFieldStress`).

4.2 P2 — Mesh stitching

Two separately-meshed half-plates merged at the seam by the coincident-node merge solve as one continuous plate with the *exact* uniaxial answer in every element, and reactions balance (test `Mesher_MergeCoincidentNodes_StitchesTwoSurfacesIntoContinuousPlate`).

4.3 P3 — Stringer load transfer (detached bar + fastener springs)

A skin plate with a detached bar along its top edge, fastened by springs at every coincident node pair. A 1000 lb axial load on the bar's free end transfers fully through the fasteners into the skin: $\Sigma R_x = -1000$ to 4 decimals, with the end bar carrying the force (test `DetachedBar_FastenedBySprings_TransfersLoad`
Model file: `stringer-load-transfer.json`.

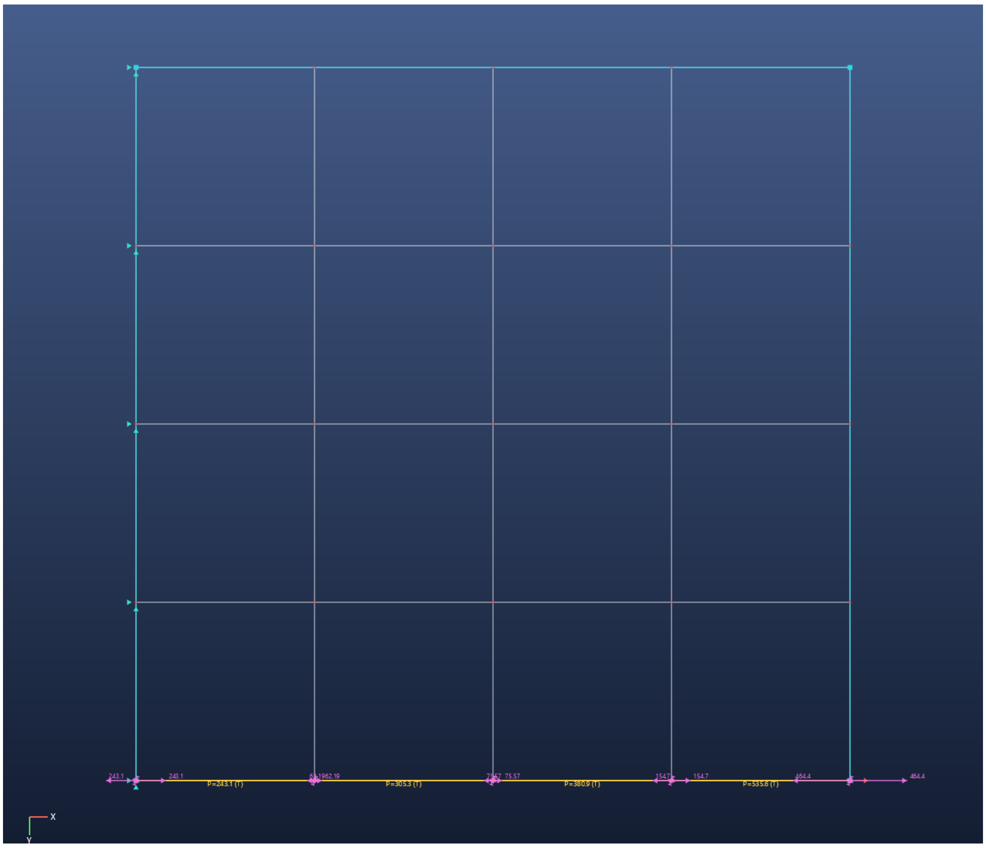


Figure 4.1: P3: detached stringer (yellow) on the skin, fastener springs (coil glyphs) and spring shear-force vectors after the solve.

4.4 P4 — Webtool sample model

A model exported from the companion webtool (curved edge, mixed BCs, bars) loads, solves, and satisfies equilibrium; the JSON format round-trips (tests `SampleModel_LoadsAndSolves`, `WebtoolJson_RoundTri`

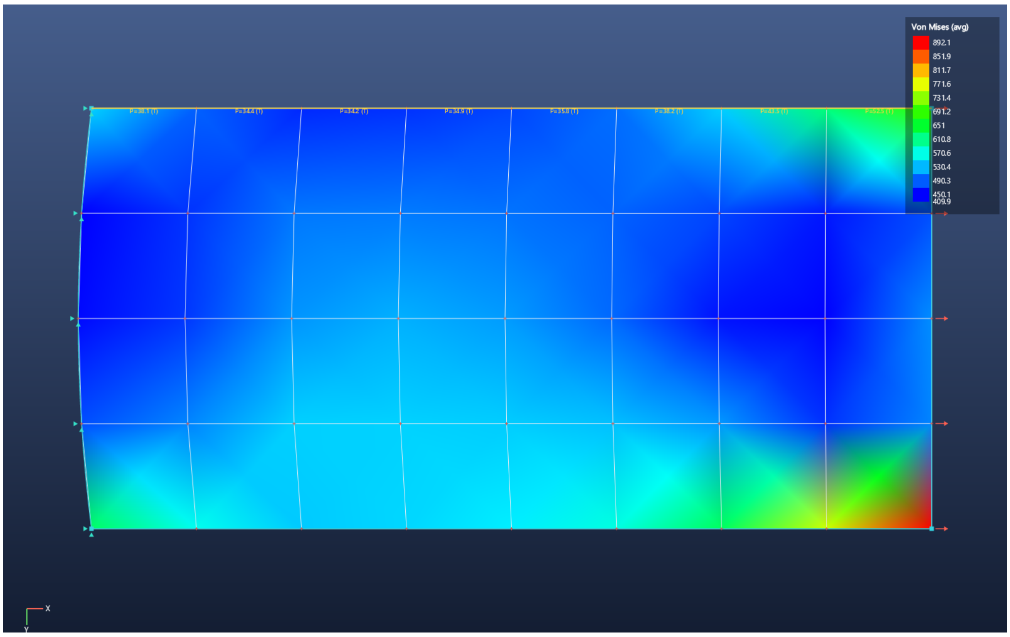


Figure 4.2: P4: curved-edge sample model, von Mises contour.

Chapter 5

External standard benchmarks (NAFEMS & SFM/AFNOR)

The cases in this chapter are recognised industry verification benchmarks with externally published reference solutions — the kind of independent corroboration expected in a damage-tolerance substantiation. Two suites are used:

- **NAFEMS**, *The Standard NAFEMS Benchmarks* (Glasgow: NAFEMS, Rev. 3, 1990) — the widely-cited finite-element verification set.
- **SFM/AFNOR**: Société Française des Mécaniciens, *Guide de validation des progiciels de calcul de structures* (Paris: Afnor Technique, 1990), the SSL series.

Reference values are quoted from those sources as tabulated in the *NX Nastran 9 Verification Manual* (Siemens PLM Software, 2013), which reproduces both suites. Each case is an automated test (Appendix A) and every figure is rendered from the committed model.

5.1 E1 — NAFEMS LE1 elliptic membrane

Quarter elliptic membrane, plane stress: inner edge $(x/2)^2 + y^2 = 1$, outer edge $(x/3.25)^2 + (y/2.75)^2 = 1$; $E = 210\,\mathrm{GPa}$, $\nu = 0.3$, $t = 0.1$; uniform 10 MPa outward pressure on the outer edge; symmetry ($u_x = 0$ on $x = 0$, $u_y = 0$ on $y = 0$). The reference quantity is the tangential stress σ_{yy} at point $D = (2, 0)$.

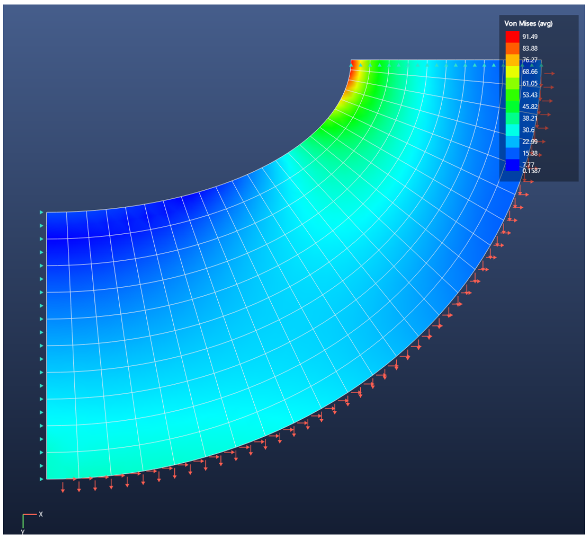


Figure 5.1: E1: LE1 elliptic membrane, von Mises field on the structured Quad8 mesh (boundary nodes lie exactly on the ellipses).

Quantity	NAFEMS bench	FEA Membranes (Quad8)	Error	NX CQUAD8
σ_{yy} at D (MPa)	92.7	92.61	$-0.1\,\%$	89.9 ($-3.0\,\%$)

Table 5.1: E1 results (test `External_NAFEMS_LE1_EllipticMembrane`). The structured mesh with nodes on the exact ellipse reaches the benchmark to within 0.1 % — closer than the coarser parabolic-quad mesh tabulated by NX.

5.2 E2 — SFM/AFNOR SSLP01 plane shear panel

A $48 \times 12 \times 1$ mm plate (plane stress), clamped at $x = 0$, with a parabolic shear traction of resultant 40 N on the free end $x = 48$; $E = 30$ GPa, $\nu = 0.25$. The reference quantity is the in-plane tip displacement u_y.

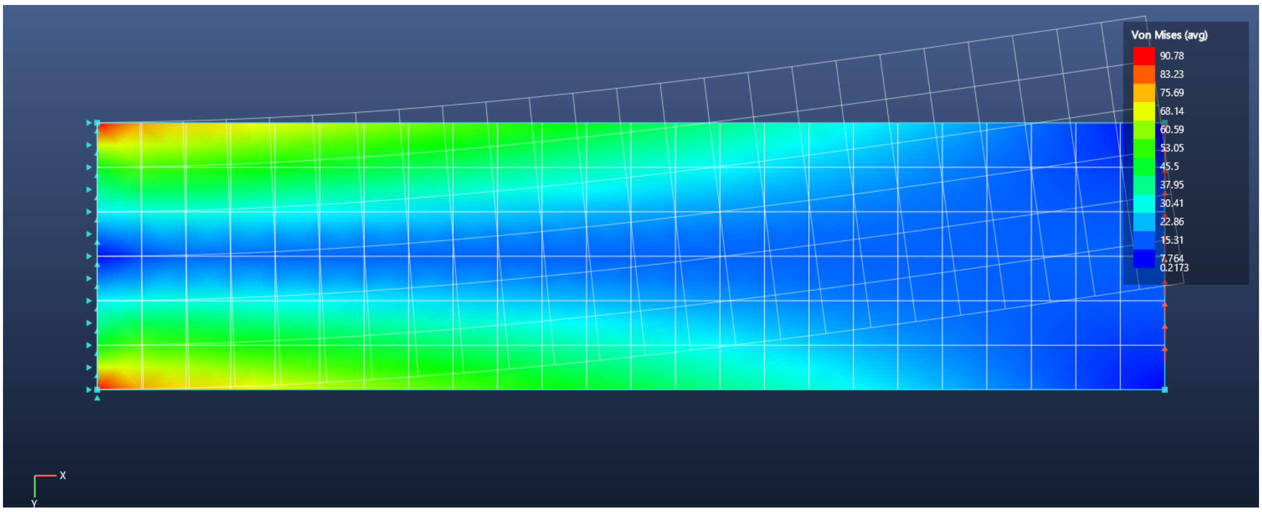


Figure 5.2: E2: SSLP01 deep cantilever under end shear, von Mises field with the deformed shape overlaid.

Quantity	AFNOR bench	FEA Membranes (Quad8)	Timoshenko	NX CQUAD4
u_y at tip (mm)	0.3413	0.3559	0.357	0.3408

Table 5.2: E2 results (test `External_SFM_SSLP01_PlaneShearCantilever`).

The AFNOR bench $0.3413 = PL^3/3EI$ is the slender-beam (bending-only) value. The plane-stress Quad8 additionally carries transverse shear and so returns 0.3559 mm, in agreement with the Timoshenko beam-plus-shear value 0.357 mm for this $h/L = 1/4$ section. This is a positive result: it confirms the Quad8 does *not* shear-lock; a locking bilinear element instead stiffens back toward the no-shear value (NX's CQUAD4 gives 0.3408).

5.3 E3 — SFM/AFNOR SSLP02 plate with a circular hole

The damage-tolerance centrepiece. A quarter plate 100×100 mm with a $r = 10$ mm circular hole, plane stress, $E = 30$ GPa, $\nu = 0.25$, remote tension $\sigma = 25$ MPa; the hole-edge hoop stress is the classic open-hole stress concentration ($K_t = 3$ for an infinite plate). The structured ring mesh is graded toward the hole.

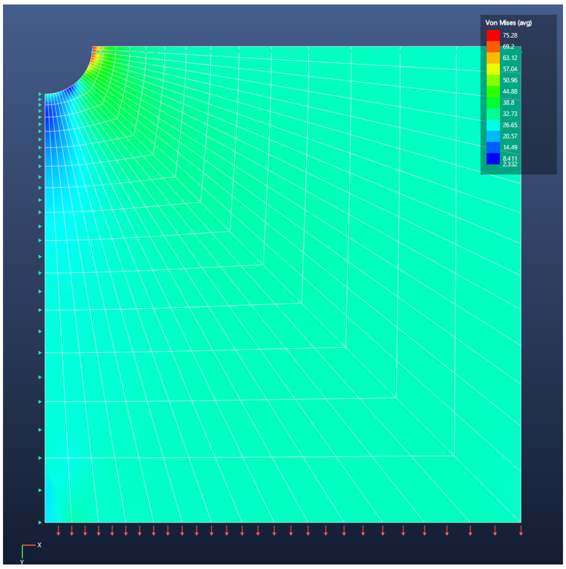


Figure 5.3: E3: SSLP02 quarter plate with a circular hole, von Mises field (legend capped at 80 MPa) showing the stress concentration at the hole edge.

Quantity	AFNOR bench	FEA Membranes	Error	NX CQUAD4
$\sigma_{\theta\theta}$ at θ=0° (MPa)	75	76.2	+1.6 %	75.3
$\sigma_{\theta\theta}$ at θ=90° (MPa)	−25	−26.2	+4.6 %	−24.5
$K_t = \sigma_{\theta\theta}/\sigma$	3.00	3.05	+1.6 %	—

Table 5.3: E3 results (test `External_SFM_SSLP02_PlateWithHole_StressConcentration`). The small overshoot is the finite-width effect ($W/d = 5$) plus discretisation; the hole-edge $K_t \approx 3$ is recovered directly. Open-hole K_t underlies fastener-hole and cutout crack-initiation and net-section/bypass assessment.

5.4 E4 — SFM/AFNOR SSLL11 articulated truss

A four-rod plane truss (bar elements): nodes $A(0,0)$, $B(1,0)$, $C(0.5,0.5)$, $D(2,1)$; rods AC, CB ($A = 2 \times 10^{-4}$), CD, BD ($A = 1 \times 10^{-4}$); $E = 2.1 \times 10^{11}$; A, B pinned; 9.81 kN at D in $-y$.

Quantity	FEA Membranes	Reference
Bar force CD (kN)	+15.51 (T)	— (statically determinate)
Bar force BD (kN)	−20.81 (C)	— (statically determinate)
u_D (mm)	(3.25, −5.23)	(3.479, −5.601)

Table 5.4: E4 results (test `External_SFM_SSLL11_ArticulatedTruss`).

The bar forces are statically determinate and reproduce a hand equilibrium/virtual-work solution exactly. The displacements match the exact analytical (virtual-work) solution for this geometry ($u_{D,y} = -5.23\,\text{mm}$, verified independently); they sit $\approx 7\,\%$ below the manual's tabulation, whose exact nodal coordinates (the support spacing / overall scale) are not deducible from the published member lengths. The displacement *direction* ($u_{D,x}/u_{D,y} = -0.621$) matches the reference exactly, confirming the geometry.

5.5 Scope: fracture benchmarks

Neither the NAFEMS nor the SFM/AFNOR linear-statics suites reproduced here include a stress-intensity-factor benchmark. The software's crack/SIF capability — the core of damage tolerance — is validated separately against handbook solutions in Chapter 6 (single-edge-notch vs Gross & Brown; centre crack vs Feddersen). Recommended companion references for that capability are Tada, Paris & Irwin (*Stress Analysis of Cracks Handbook*), Murakami (*Stress Intensity Factors Handbook*) and Rooke & Cartwright (*Compendium of SIF*).

Chapter 6

Stress intensity factors

Method per Section 1.5: quarter-point Quad8 crack tips, domain interaction integral (primary), displacement correlation (cross-check). Model files: `crack-sent-benchmark.json`, `crack-cct-benchmark.json`.

6.1 K1 — Single-edge-notched tension (SENT)

$W = 10$, $a/W = 0.3$, $H/W = 1.5$ per side, remote tension $\sigma = 100$ psi applied as consistent tractions, 40×120 Quad8 mesh ($L_{tip}/a = 0.083$). Handbook (Gross & Brown): $K_I = Y\sigma\sqrt{\pi a}$, $Y = 1.122 - 0.231r + 10.55r^2 - 21.71r^3 + 30.382r^4 = 1.6625$ at $r = 0.3$, giving $K_I = 510.3$.

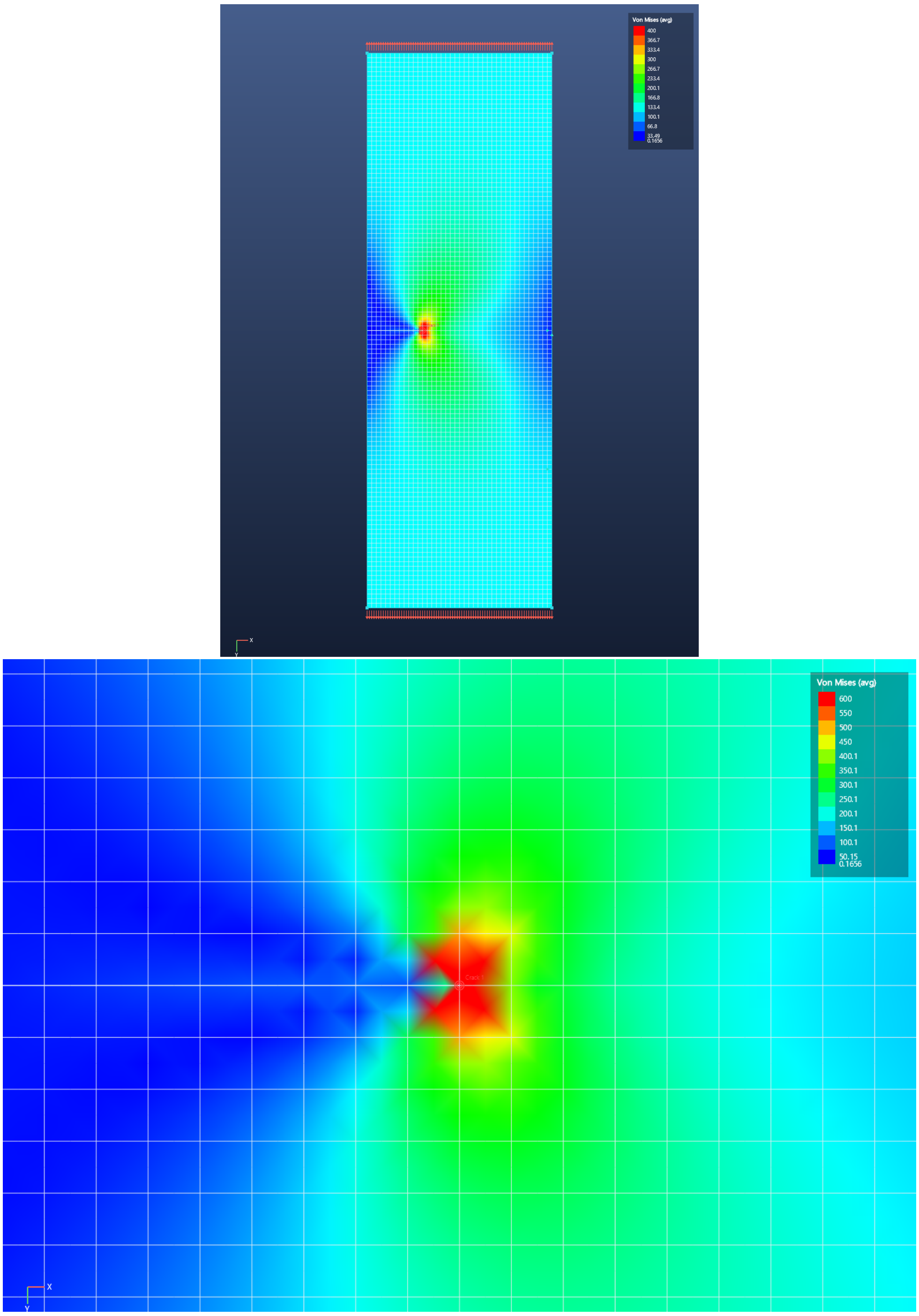


Figure 6.1: K1: SENT von Mises field (legend capped for display) and tip close-up showing the characteristic lobed field and unloaded crack wake.

Quantity	Handbook	FEA Membranes	Error	NASTRAN CQUAD8R
K_I (interaction integral)	510.3	509.7	$-0.1\,\%$	—
K_I (displacement correlation)	510.3	534.6	$+4.8\,\%$	—
K_{II}	0	≈ 0	$< 2\,\%$ of K_I	—

Table 6.1: K1 results (test `Crack_EdgeCrackedPlate_MatchesHandbookK1`).

6.2 K2 — Centre-cracked tension (CCT), both tips

Plate 20×40, half-crack $a = 4$ ($a/W = 0.4$), both tips created in one command. Handbook (Feddersen): $K_I = \sigma\sqrt{\pi a}\,\sec(\pi a/2W) = 394.1$.

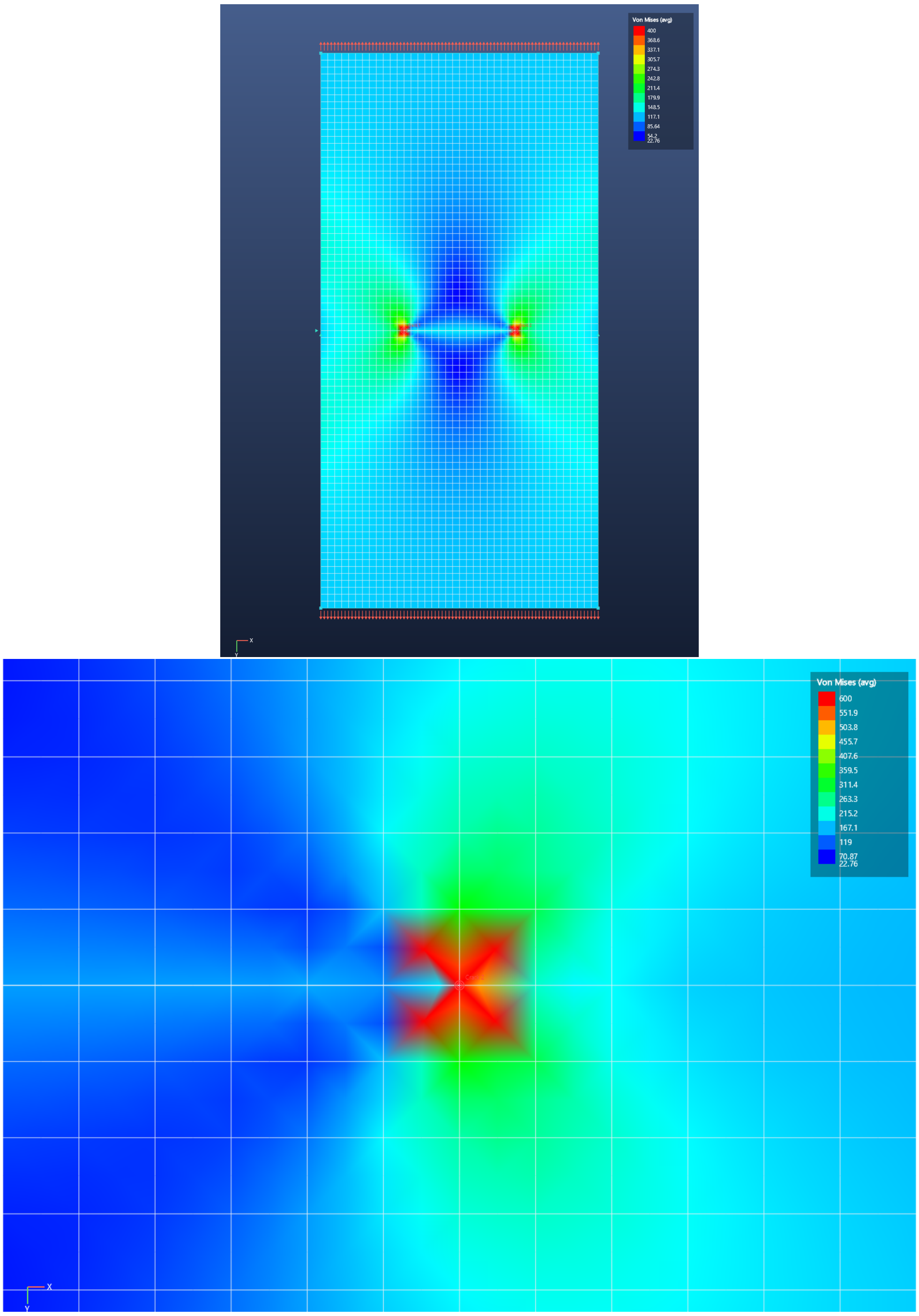


Figure 6.2: K2: CCT von Mises field and right-tip close-up.

Quantity	Handbook	FEA Membranes	Error	NASTRAN CQUAD8R
K_I (interaction integral)	394.1	393.7	$-0.1\,\%$	—
K_I (displacement correlation)	394.1	418.9	$+6.3\,\%$	—
K_{II}	0	≈ 0	$< 2\,\%$ of K_I	—
Tip symmetry	equal	equal	1 decimal	—

Table 6.2: K2 results (test `Crack_CentreCrackedPlate_MatchesHandbookK1_BothTips`).

6.3 K3 — Domain independence

The defining correctness property of a domain integral: the same SENT solution evaluated over integration annuli of 2, 3, 4 and 6 tip-element lengths agrees within **1 %** of the mean (test `Crack_InteractionIntegra`

6.4 Method accuracy statement

The interaction integral measures $-0.1\,\%$ on both handbook benchmarks and is domain-independent to $< 1\,\%$. The displacement-correlation cross-check carries the known $\approx 5\,\%$ bias of non-collapsed rectangular quarter-point arrangements ($+5$ to $+6\,\%$ with the production reduced-integration Q8). In application work, a large divergence between the primary K and the DCT value should prompt inspection of the tip mesh.

Chapter 7

Diagnostics and model-integrity verification

- **Orphan nodes**: a pre-solve check names any node with no attached stiffness instead of failing opaquely.
- **Under-constrained models**: the sparse LU factorisation does not necessarily fail on a singular matrix; every solution is therefore verified against $\|Ku - f\| \leq 10^{-6}\|f\|$ and rigid-body mechanisms are rejected with a clear message. (This guard exists because testing caught the factorisation returning silent zeros.)
- **Editing integrity** (16 automated cases): re-meshing fully replaces meshes; deletion of nodes/elements/surfa cascades to everything referencing them; coincident-node merging never destroys spring-joined fastener pairs nor heals crack faces; RBE2 and crack records are remapped or pruned through all operations; the webtool JSON format round-trips.

Chapter 8

NASTRAN comparison programme

The grey dashes throughout this manual are reserved for MSC NASTRAN results using CQUAD8(R) elements on identical meshes. Recommended procedure per case:

1. Open the benchmark model (`validation/*.json`) in FEA Membranes and export the results CSV (*File → Export Results as CSV*); the NODES section gives the exact node coordinates for the NASTRAN deck, and the BC columns give the constraint/load sets.
2. Reproduce mesh, constraints and consistent loads in NASTRAN (CQUAD8 + PSHELL membrane-only, or CQUAD8R where available); for the crack cases use the same quarter-point node positions (the CSV coordinates already include the quarter-point shifts).
3. Record tip deflections / stresses / K values in the comparison columns; note the NASTRAN version, element keyword and any parameter settings (e.g. K6ROT, SNORM defaults are irrelevant for membranes) beside each table.

Appendix A

Reproduction

A.1 Run the automated suite

```
dotnet test tests/FeaCore.Tests/FeaCore.Tests.csproj
```

46/46 cases pass at this revision. The suite also regenerates every benchmark model into `validation/`.

A.2 Regenerate the figures

```
FeaMembranes.exe --render <model.json> <out.png>
    [--contour vm|sx|sy|sxy|none] [--deformed] [--no-nodes]
    [--vectors springs|reactions|both] [--cmax <value>]
    [--zoom <cx> <cy> <halfwidth>] [--width N] [--height N]
```

The exact commands used for the figures in this manual are recorded in `validation/img/render-commands.ps1`.

A.3 Automated test inventory

The complete case-by-case record, with per-test acceptance tolerances, is maintained in `validation/VALIDATION.` (kept current by construction: a failing case fails the build).